

Jiaqing Lu

Curriculum Vitae

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Education

- 2013 - 2018 **Doctor of Philosophy, The Ohio State University,**
Major: Electrical and Electronics Engineering, Advisor: Prof. Jin-Fa Lee.
Dissertation: Numerical Modeling and Computation of Radio Frequency Devices.
- 2010 - 2013 **Master of Science in Engineering, Shanghai Jiao Tong University,**
Major: Electronics Science and Technology, Advisor: Prof. Junfa Mao.
Thesis: Transient Simulation and Sensitivity Analysis of Transmission Lines Based on Waveform Relaxation Method.
- 2006 - 2010 **Bachelor of Science in Engineering, Shanghai Jiao Tong University,**
Major: Information Engineering, Advisor: Prof. Min Tang.
Thesis: Transient Analysis of High-Speed Interconnects.

Research Interests

- Computational Electromagnetics, Finite Element Analysis, Domain Decomposition Methods, Fast Direct Solvers and Preconditioners, Hierarchical Matrix Techniques
- Electromagnetic Compatibility, Signal Integrity
- Electromagnetic Analysis of High-Speed Microelectronics, Electromagnetic Emission and Vulnerability of Integrated Circuits

Academic & Professional Experience

- 2026 - **Associate Professor, Shanghai Jiao Tong University:**
- School of Integrated Circuits | School of Information Science and Electronic Engineering
 - State Key Laboratory of Radio-Frequency Heterogeneous Integration
- 2025 **Algorithm Engineer, NineCube Microelectronics Co., Ltd:**
- Finite element solver and EDA software development
- 2019 - 2024 **Postdoctoral Scholar, ElectroScience Laboratory, The Ohio State University:**
- Developed mixed frequency-domain electromagnetic and time-domain circuit co-simulation package for high-frequency circuit analysis

- Conducted electromagnetic signature simulations and measurements of integrated circuits regarding near-field emissions and near-field/far-field vulnerabilities
- Developed higher-order absorbing boundary condition for finite element analysis
- Developed a novel direct solver for nonconformal domain decomposition method
- Developed a hierarchical direct solver for finite element method with boundary skeletonization
- Developed a preconditioner for embedded domain decomposition method targeting spectral radius reduction
- Developed a multi-level preconditioner for nonconformal domain decomposition method
- Software & GUI development for domain decomposition methods, resulting in a software package used for antenna design at Singapore DSO National Laboratory

2013 - 2018 **Research Associate, ElectroScience Laboratory, The Ohio State University:**

- Developed an embedded domain decomposition method for electromagnetic modeling and design, successfully applied to areas of antenna design, electromagnetic compatibility, and integrated circuits
- Integrated discontinuous Galerkin time domain method with SPICE & IBIS circuit model simulations and developed an electromagnetic-circuit co-simulation package for high-speed circuit analysis

2016 **Intern, HFSS Development Team, Ansys, Inc.**

- Developed HFSS Test Suite for Ansys Electronics Desktop

2010 - 2013 **Graduate Student, Key Laboratory of China for Research of Design and Electromagnetic Compatibility of High Speed Electronic Systems, Shanghai Jiao Tong University:**

- Code development integrating finite difference time-domain method and waveform relaxation method for electromagnetic interference simulations of printed circuit boards
- Code development for analyzing electro-thermal behavior and parametric thermal sensitivity of on-chip interconnects
- Through-silicon via modeling and analysis in electronic packaging systems in cooperation with Huawei Technologies

Projects & Grants

2027 - 2030 **Embedded Domain Decomposition Method for Multiscale Electromagnetic Systems,
Funded by National Natural Science Foundation of China.**

Role: Principal Investigator

- 2026 - 2028 **Efficient Solvers for Large-Scale Multiphysics Domain Decomposition Systems, Funded by State Key Laboratory of Radio-Frequency Heterogeneous Integration, China.**
Role: Principal Investigator
- 2019 - 2024 **CYAN: Enabling Cyber Defense in Analog and Mixed Signal Domains, Funded by Air Force Research Laboratory, USA.**
Role: Primary contributor for a task – Electromagnetic Signature Simulations for AMS Electronics and Vulnerability Analysis;
Led simulation team, measurement team, and IC design team for the task;
Developed electromagnetic-circuit co-simulation package for integrated circuits;
Authentication method development for IC chips through electromagnetic emissions.
- 2020 - 2024 **STAMP: A Holistic Backward/Forward Trust Framework for Protecting Microelectronics Throughout Lifecycle, Funded by Air Force Research Laboratory, USA.**
Role: Responsible for setting up simulation and measurement platforms for the electromagnetic vulnerability of microelectronics designed/manufactured by IC team.
- 2010 - 2023 **Advanced Computational HF Techniques, Funded by Ansys, Inc., USA**
Role: Primary contributor since 2016;
Communicated with HFSS development team and developed novel algorithms;
Tasks include:
 - Advanced Direct Solver for FEM and DDM;
 - Higher-Order Absorbing Boundary Conditions;
 - Incorporate Generalized Impedance Boundary Condition into FEM formulation;
 - Advanced CEM Algorithms for Signal Integrity Analyses in ICs and Packages – Chimera Overset-Grid Methods for Signal Integrity Analyses.
- 2018 - 2024 **Development of integral equation discontinuous Galerkin method, finite element method, and domain decomposition method for high-quality factor cavity problems, Funded by Sandia National Laboratories, USA.**
Role: Algorithm development for high-quality factor cavity problems using finite element method and domain decomposition method.
- 2017 - 2020 **Design code development for periodic/apperiodic electromagnetic structures, Funded by DSO National Laboratories, Singapore.**
Role: Primary contributor for the project;
Developed simulation package for periodic electromagnetic structures using embedded domain decomposition method;
Developed a GUI for embedded and nonconformal domain decomposition methods.

2013 - 2015 **Electromagnetic wave couplings,
Funded by DSO National Laboratories, Singapore.**

Role: Primary contributor for the project;
Developed electromagnetic-circuit co-simulation package for circuit board;
Built interface between discontinuous Galerkin method and SPICE simulator;
Built interface between discontinuous Galerkin method and behavioral modeling.

Awards & Honors

- 2025 **Overseas Excellent Young Scientists Fund Program of China**
- 2025 **Shanghai Magnolia Talent Program**
- 2025 Progress in Electromagnetics Research Symposium (PIERS), **Young Scientist Award**
- 2025 IEEE MTT-S International Microwave Symposium (IMS), **TOP 50 Paper Award**
- 2020 IEEE International Symposium on Electromagnetic Compatibility, Signal Integrity and Power Integrity, **Best EMC Symposium Paper Award – Honorable Mention**
- 2018 IEEE International Symposium on Electromagnetic Compatibility, Signal Integrity and Power Integrity, **Best Symposium Paper Finalist**
- 2018 IEEE International Symposium on Electromagnetic Compatibility, Signal Integrity and Power Integrity, **Best Student Paper Finalist**

Publications

Journal Articles

- 2025 ■ **Jiaqing Lu**, "Electromagnetic emission simulation of radio frequency circuits using direct domain decomposition solver," *IEEE Microw. Wireless Technol. Lett.*, vol. 35, no. 6, pp. 768–771, Jun. 2025.
- 2024 ■ **Jiaqing Lu**, "An improved absorbing boundary condition for finite element analysis," *IEEE Trans. Antennas Propag.*, vol. 72, no. 10, pp. 8109–8114, Oct. 2024.
- **Jiaqing Lu**, Yagmur Ozturk, Wentao Bao, Jingyue Zhang, Shane Smith, Kubilay Sertel, and Jin-Fa Lee, "Vulnerability assessment of integrated circuits through electromagnetic-circuit cosimulation," *IEEE Trans. Electromagn. Compat.*, vol. 66, no. 5, pp. 1525–1541, Oct. 2024.
- **Jiaqing Lu**, "Convergence enhancement for embedded domain decomposition method through spectral radius reduction," *IEEE Trans. Microw. Theory Techn.*, vol. 72, no. 10, pp. 5745–5758, Oct. 2024.
- **Jiaqing Lu**, "A low-rank direct solver for nonconformal domain decomposition methods," *IEEE Trans. Antennas Propag.*, vol. 72, no. 8, pp. 6663–6677, Aug. 2024.

- **Jiaqing Lu**, “A preconditioning approach for the domain decomposition simulation of high-speed circuits,” *IEEE Trans. Microw. Theory Techn.*, vol. 72, no. 6, pp. 3345–3360, Jun. 2024
- **Jiaqing Lu** and Jin-Fa Lee, “A compression scheme for domain decomposition method in solving electromagnetic problems,” *J. Comput. Phys.*, vol. 503, Art. no. 112824, Apr. 2024
- **Jiaqing Lu**, Yagmur Ozturk, Cem Kesici, Shane Smith, Kubilay Sertel, and Jin-Fa Lee, “Full-wave modeling and experimental characterization of electromagnetic emissions from printed circuit boards,” *IEEE Trans. Microw. Theory Techn.*, vol. 72, no. 4, pp. 2427–2441, Apr. 2024
- 2023 ▪ **Jiaqing Lu** and Jin-Fa Lee, “Convergence improvement for domain decomposition method through corner treatment,” *IEEE Antennas Wireless Propag. Lett.*, vol. 22, no. 12, pp. 2965–2969, Dec. 2023.
- 2019 ▪ **Jiaqing Lu**, Yongpin Chen, Dongwei Li, and Jin-Fa Lee, “An embedded domain decomposition method for electromagnetic modeling and design,” *IEEE Trans. Antennas Propag.*, vol. 67, no. 1, pp. 309–323, Jan. 2019
- 2018 ▪ **Jiaqing Lu** and Jin-Fa Lee, “Signal integrity analysis of integrated circuits by using embedded domain decomposition method,” *IEEE Trans. Microw. Theory Techn.*, vol. 66, no. 12, pp. 5369–5381, Dec. 2018
- 2014 ▪ Guangcao Fu, Min Tang, **Jiaqing Lu**, Junfa Mao, “Thermal sensitivity analysis of on-chip interconnects,” *Electron. Lett.*, vol. 50, no. 11, pp. 797–799, May. 2014.
- 2013 ▪ Min Tang, **Jiaqing Lu**, Junfa Mao, Lijun Jiang, “A systematic electromagnetic-circuit method for EMI analysis of coupled interconnects on dispersive dielectrics,” *IEEE Trans. Microw. Theory Techn.*, vol. 61, no. 1, pp. 1–13, Jan. 2013.

Conference Papers & Talks

- 2026 ▪ **Jiaqing Lu**, “A Low-Rank Direct Solver for Sparse Matrices Associated with FEM and DDM Systems,” in *International Applied Computational Electromagnetics Society Symposium (ACES-China)*, Qingdao, China, Aug. 2026.
- **Jiaqing Lu** and Min Tang, “Formulation and Performance Analysis of Embedded Domain Decomposition Method for Electromagnetic Simulations,” in *International Applied Computational Electromagnetics Society Symposium (ACES-China)*, Dingdao, China, Aug. 2026.
- **Jiaqing Lu** and Min Tang, “Formulation and Performance Analysis of Embedded Domain Decomposition Method for Electromagnetic Simulations,” in *International Conference on Microwave and Microelectronics of Communications (ICMMC)*, Beijing, China, Aug. 2026.

- **Jiaqing Lu** and Min Tang, "Domain Decomposition Modeling and EM-Circuit Co-Simulation for Electromagnetic Compatibility Analysis of Radio-Frequency Circuits," in *Conference on Microwave, Terahertz, and Electromagnetic Compatibility Technologies (MMW-THZ-EMC)*, Dalian, China, Aug. 2026.
- **Jiaqing Lu**, "Electromagnetic Signature Simulation and Vulnerability Analysis of Radio Frequency Circuits," in *The 9th International Symposium on Electromagnetic Compatibility (ISEMC)*, Xi'an, China, Apr. 2026.
- 2025 ▪ **Jiaqing Lu** and Jin-Fa Lee, "A Direct Domain Decomposition Solver with Multi-level Skeletonization and Higher-order Absorbing Boundary Condition," in *2025 Progress in Electromagnetics Research Symposium (PIERS)*, Abu Dhabi, UAE, May. 2025.
- 2022 ▪ **Jiaqing Lu**, Yagmur Ozturk, Cem Kesici, Kubilay Sertel, Shane Smith, and Jin-Fa Lee, "Electromagnetic Signature Simulations for AMS Electronics," in *2022 Trusted and Assured Microelectronics (TAME) Forum*, Columbus, OH, USA, Sep. 2022.
- **Jiaqing Lu**, Yagmur Ozturk, Cem Kesici, Jingyue Zhang, Shane Smith, Kubilay Sertel, and Jin-Fa Lee, "Electromagnetic Vulnerability Analysis of AMS Electronics," in *2022 Trusted and Assured Microelectronics (TAME) Forum*, Columbus, OH, USA, Sep. 2022.
- 2021 ▪ **Jiaqing Lu**, Yagmur Ozturk, Cem Kesici, Shane Smith, Kubilay Sertel, and Jin-Fa Lee, "Counterfeit chip detection via electromagnetic emissions from host circuit boards," in *IEEE Antennas Propag. Soc. Int. Symp. (APSURSI)*, Singapore, Dec. 2021.
- 2020 ▪ **Jiaqing Lu** and Jin-Fa Lee, "Approximate direct inverse for embedded domain decomposition method," in *Proc. IEEE Int. Symp. Electromagn. Compat. Signal/Power Integrity*, Reno, NV, USA, Aug. 2020, pp. 45-49.
- **Jiaqing Lu** and Jin-Fa Lee, "Approximate direct inverse for embedded domain decomposition method," in *IEEE Antennas Propag. Soc. Int. Symp. (APSURSI)*, Montreal, QC, Canada, Jul. 2020.
- 2019 ▪ **Jiaqing Lu** and Jin-Fa Lee, "Complex antenna simulation by using embedded domain decomposition," in *IEEE Antennas Propag. Soc. Int. Symp. (APSURSI)*, Atlanta, GA, USA, Jul. 2019.
- **Jiaqing Lu**, Dongwei Li, Wentao Bao and Jin-Fa Lee, "Embedded domain decomposition approach for analyzing complex antenna structures," in *Proc. IEEE Int. Conf. Numer. Electromag. Multiphys. Modeling Optim. (NEMO)*, Boston, MA, USA, May 2019, pp. 1-4.
- 2018 ▪ **Jiaqing Lu** and Jin-Fa Lee, "Electromagnetic compatibility analysis using embedded domain decomposition method," in *Proc. IEEE Int. Symp. Electromagn. Compat. Signal/Power Integrity*, Long Beach, CA, USA, Jul. 2018, pp. 187-192.

- **Jiaqing Lu** and Jin-Fa Lee, "An embedded domain decomposition method for time-harmonic electromagnetic problems," in *Proc. IEEE MTT-S Int. Microw. Symp. Dig.*, Philadelphia, PA, USA, Jun. 2018, pp. 216-219.
- 2013 ▪ Guangcao Fu, **Jiaqing Lu**, Min Tang, Junfa Mao, "A joint-diagonalization technique for fast simulation of massive strongly coupled cables," in *Proc. Electrical Design Adv. Packag. Systems Symp. Dig.*, Dec. 2013.
- 2012 ▪ **Jiaqing Lu**, Min Tang, Junfa Mao, "Fast crosstalk analysis of multi-walled carbon nanotube interconnects," in *Proc. Electrical Design Adv. Packag. Systems Symp. Dig.*, Dec. 2012.
- Min Tang, **Jiaqing Lu**, Junfa Mao, "Study on equivalent single conductor model of multi-walled carbon nanotube interconnects," in *Proc. Asia-Pacific Microw. Conf. Dig.*, Dec. 2012.
 - **Jiaqing Lu**, Min Tang, Junfa Mao, "Fast simulation of nonuniform substrate temperature effects on global GSI interconnects," in *Proc. Asia-Pacific Microw. Conf. Dig.*, Dec. 2012.
- 2011 ▪ Min Tang, **Jiaqing Lu**, Junfa Mao, "EMI analysis of coupled interconnects on dispersive dielectrics," in *Proc. IEEE MTT-S Int. Microw. Symp. Dig.*, Jun. 2011.